Analysis & Prediction of Interactions in Large Social Networks

Group 37

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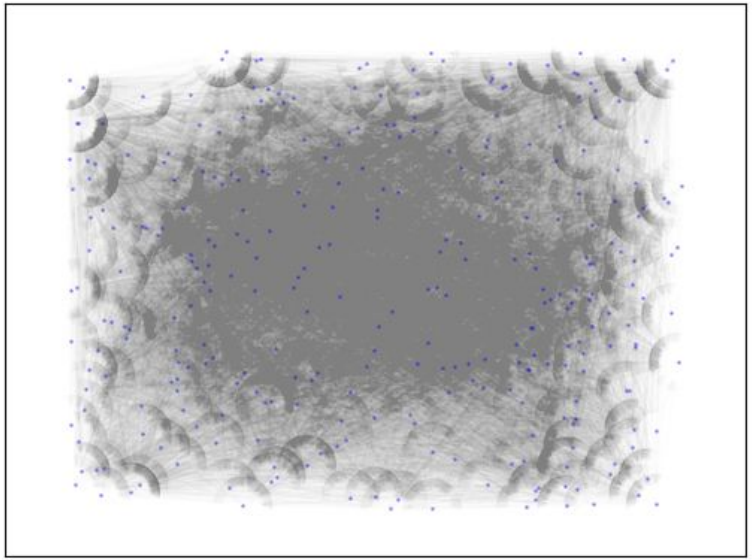
Dataset

We are using the Reddit hyperlink network. This network represents the directed connections between two subreddits (a subreddit is a community on Reddit). The network is extracted from publicly available Reddit data of 2.5 years from Jan 2014 to April 2017.

The network is directed, signed, temporal and attributed. The static network (visualization attached) has 67,180 nodes and 339,643 edges. Its density is 0.00007526 and sparsity is 0.99992474.



EDA

The network is not strongly connected.

Number of Strongly Connected Components: 45564

Number of Weakly Connected Components: 712

Largest Strongly Connected Component Size: 21432

Largest Weakly Connected Component Size: 65648

Average In/Out Degree: 5.0557

Max In-Degree: 5448 (Other subreddits have linked to this subreddit the most)

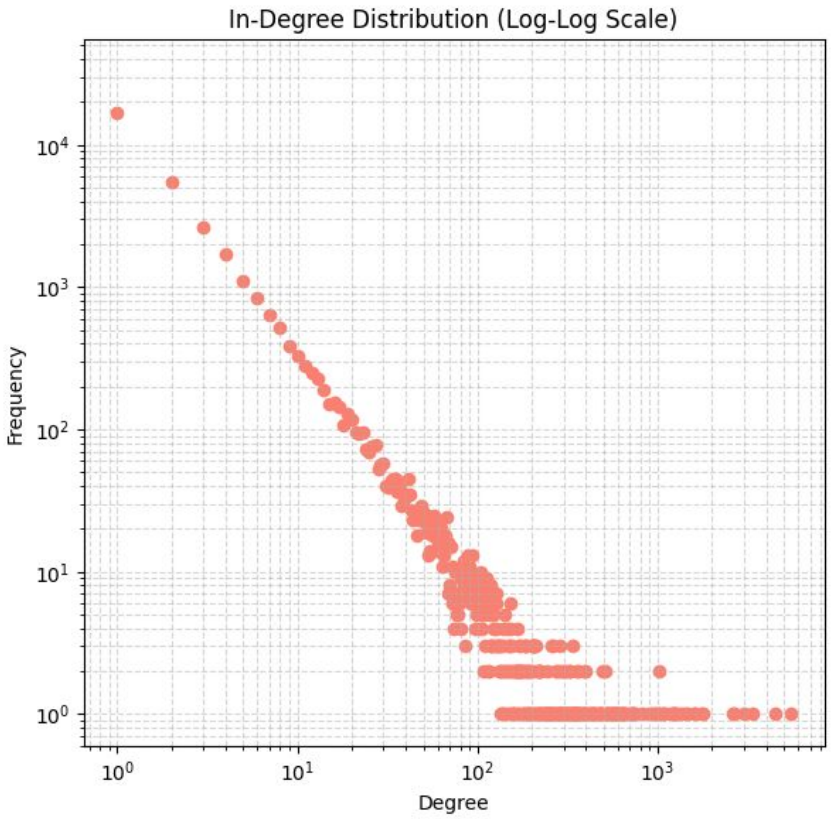
Max Out-Degree: 3111 (This subreddit has linked to other subreddits the most)

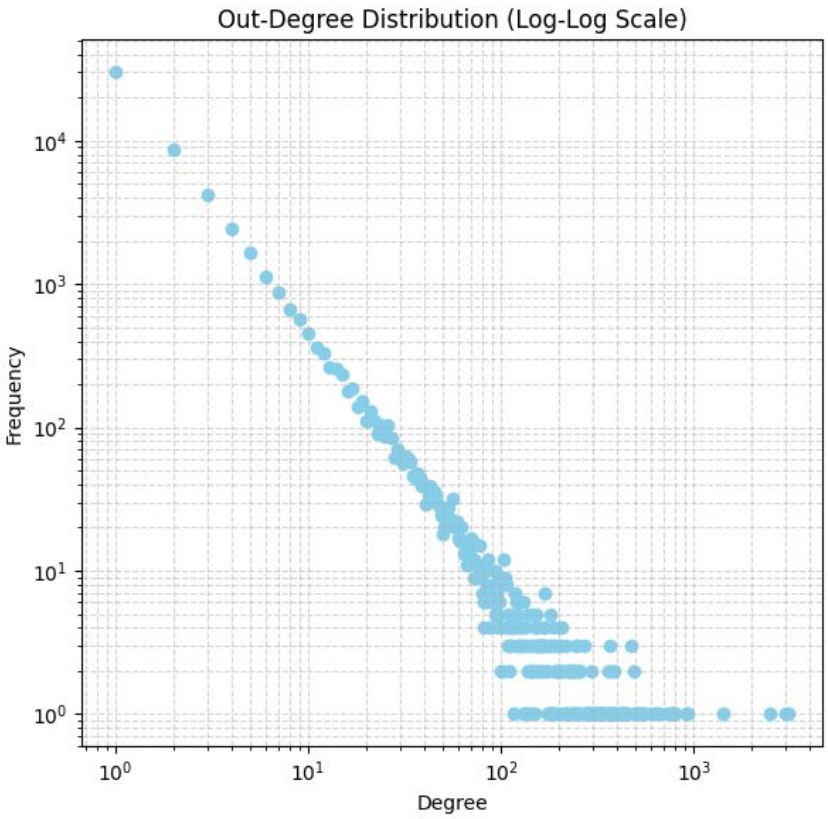
Transitivity: 0.05885

Average Clustering Coefficient: 0.16871

Degree Assortativity Coefficient: -0.10324

Degree Distribution





1) Classical Link Prediction

- Link Prediction Problem: Can we predict future social interactions (t -> t') using only the current network topology?

- Experimental Design:
  - Data Split: 80% Training (E_{old}) | 20% Testing (E_{new}).
  - Core Constraint: Evaluation limited to nodes with >= 50 interactions to ensure statistical significance.
  - Challenge: Rank all possible non-existent pairs *(u, v)* to find the likely future edges.

- Success Metric: Factor over Random tells how much better does the model perform compared to a blind guess?

Results

Nodes in Core set = 1013

$|E_{old}| = 278{,}824$, $|E_{new}| = 9{,}410$

Total Candidate Pairs = 970,696

Predictor	Hits	Factor over Random
Common Neighbors	506	5.55x
Jaccard's Coefficient	497	5.45x
Adamic/Adar	512	5.61x
Preferential Attachment	441	4.38x
Graph Distance	281	3.08x
Katz Index (=0.005)	373	4.09x
Rooted PageRank (=0.15)	317	3.48x

Conclusions

- Local measures (Adamic/Adar, Common Neighbors) outperformed Global path measures in this dataset.

- Simple neighborhood overlaps are often more predictive than complex random walks.

- This replication successfully demonstrates the core principles of the Liben-Nowell and Kleinberg paper. While the factors over random in this Reddit sample (~5x) are lower than the 40x–50x seen in the original co-authorship studies, the relative ranking of methods remains largely consistent, validating the robustness of the proximity heuristics across different types of social networks.

2) Scaling Link Prediction with Node Embeddings

- Classical methods like Jaccard:
  - Use only local neighborhood
  - Ignore semantic similarity
- Not effective for large ,complex network
- So our approach :
  - Represent each subreddit as a 300 Dim embedding
  - Convert link prediction to ML problem
- Methods used:
  - Logistic regression(embedding based)
  - Graph neural network
- Our goal is to capture structural and semantic similarity for scalable prediction

Pipeline and results

- Dataset:
  - 858k edges , 67k nodes
- Train / Test split:
  - 80% / 20% (before sampling)
- Balanced positive and negative edges
- Edge features:
  - Hadamard
  - Concatenation
  - Absolute difference
- Embedding based methods outperform structural baseline

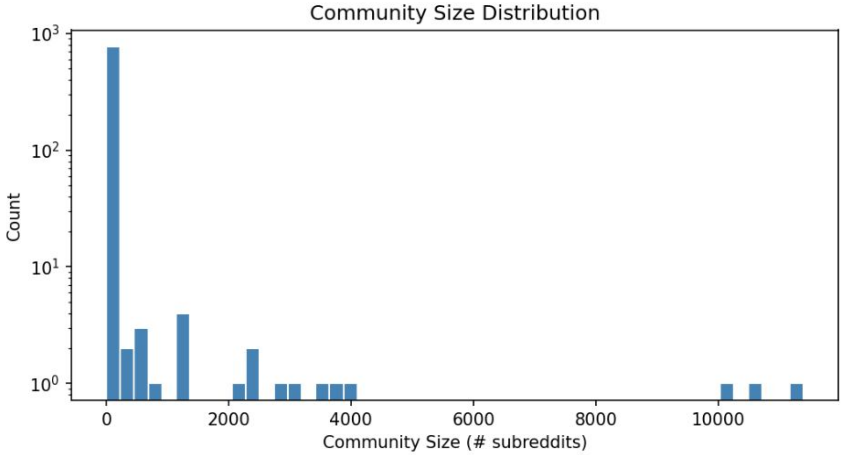
Method	AUC
Jaccard Baseline	0.9430
LR-Hadamard	0.9615
LR-Concat	0.9601
LR-AbsDiff	0.9249
GNN (held-out eval)	0.8915
Best LR vs GNN gain:	-0.0700

Insights and Interpretations

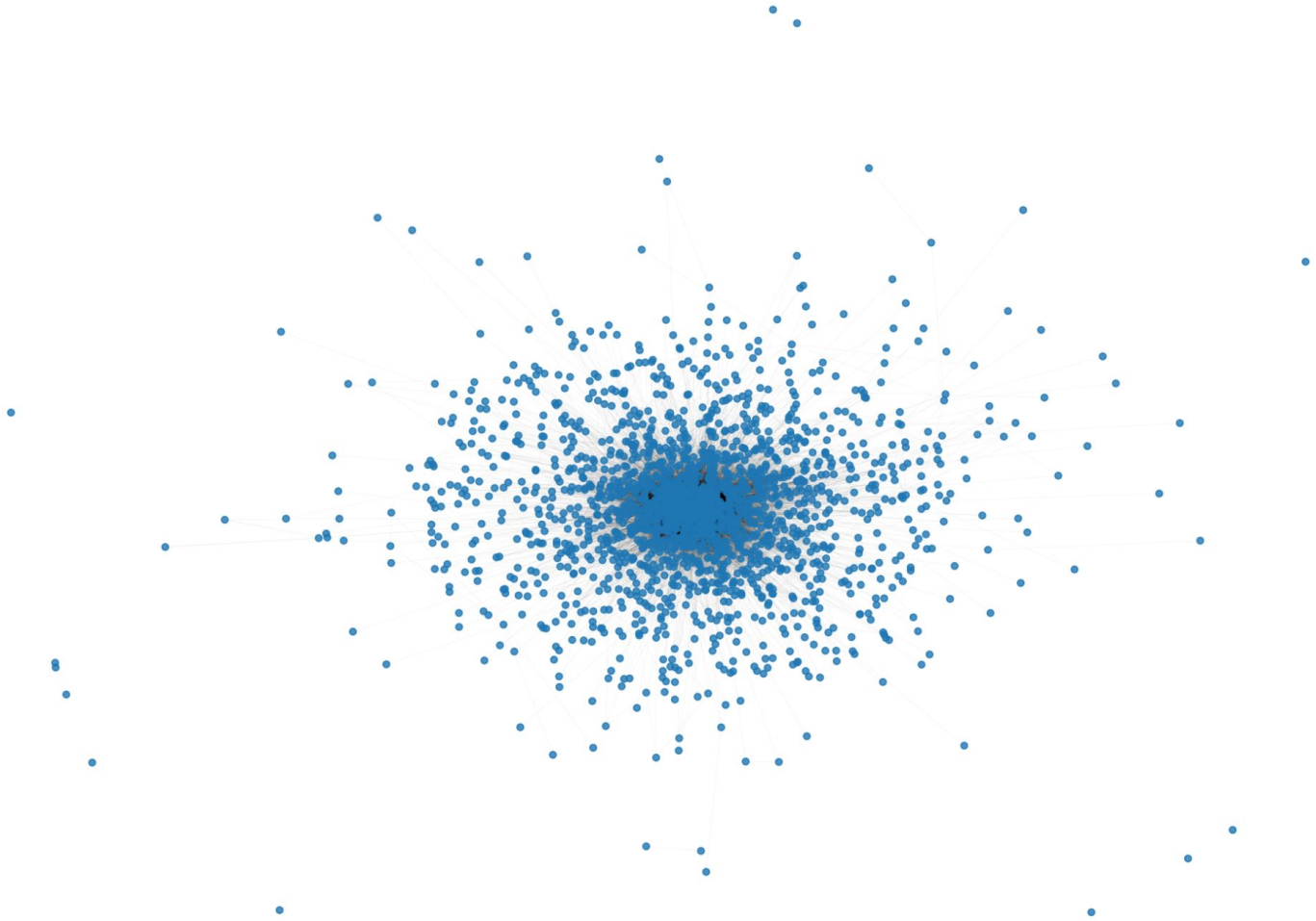
- Embeddings capture:
  - Higher order structure
  - Semantic similarity between subreddits
- Observations:
  - Hadamard = best feature
  - Concat is higher cost and no gain
  - GNN underperforms
- Link formation depends on topology and latent similarity
- Scaling link prediction with embeddings improves generalization in large networks

3) Community Detection & Analysis

- Total Communities : 789
- Modularity : 0.41 (strong structure)
- 62.4% edges are intra community
- Few large communities and small ones Indicates non random and clustered network
- Interactions are highly driven by community structure

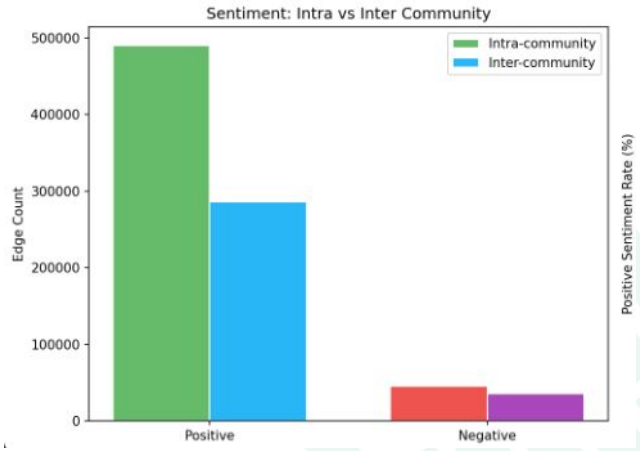


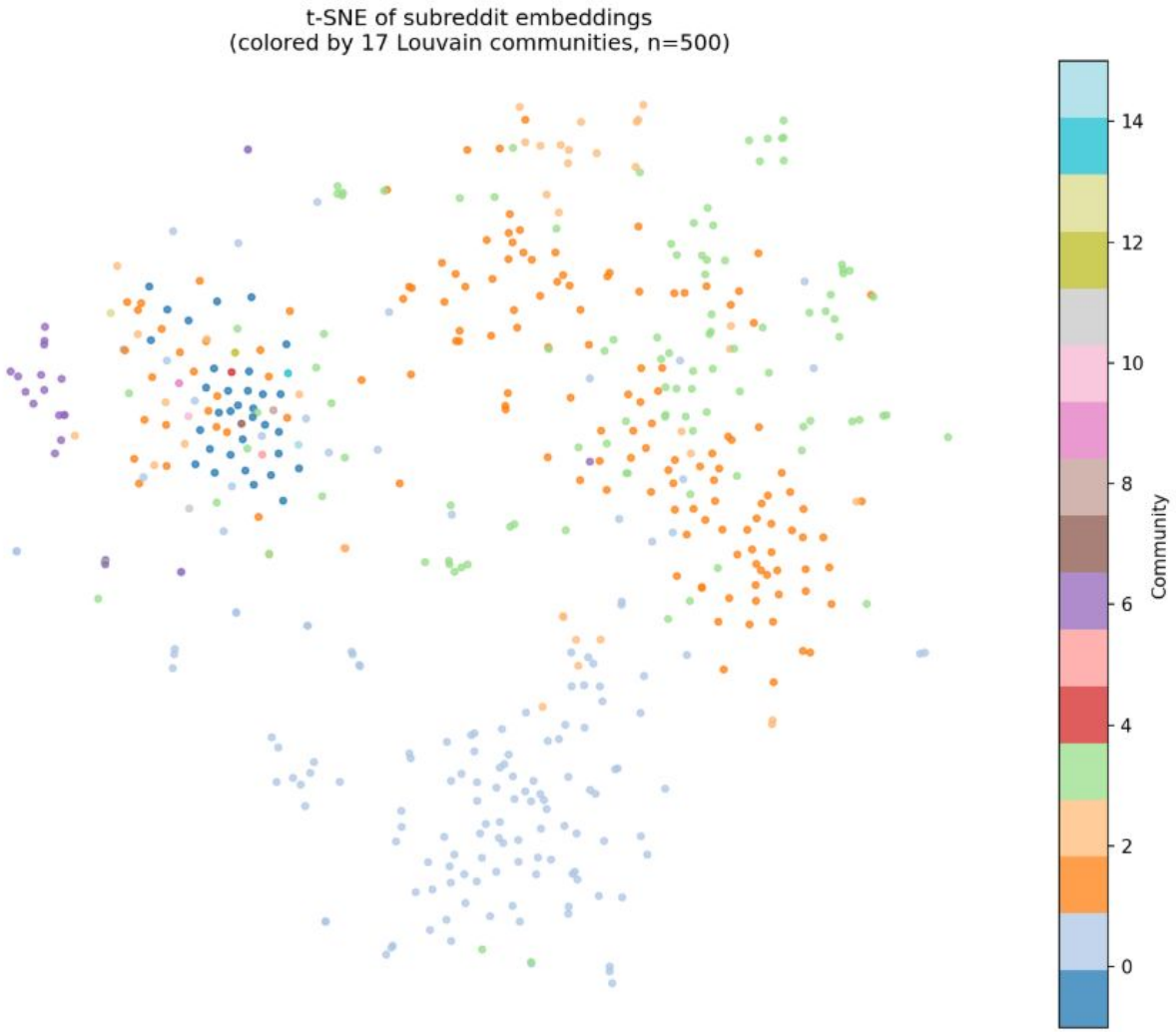
Reddit Community Structure (Top 10 Communities)



Community influence on Interaction Behaviour

- Sentiment Analysis :
  - Intra community +ive rate : 91.4%
  - Inter community +ive rate : 88.7%
- Majority interaction with community 62.4%
- Community are not just structural clusters
- They represent shared interest and aligned behaviour
- Strong evidence of emotional cohesion within communities




t-SNE of subreddit embeddings
(colored by 17 Louvain communities, n=500)
Community
0
2
4
6
8
10
12
14

4) Centrality Analysis – Identifying Influential Subreddits

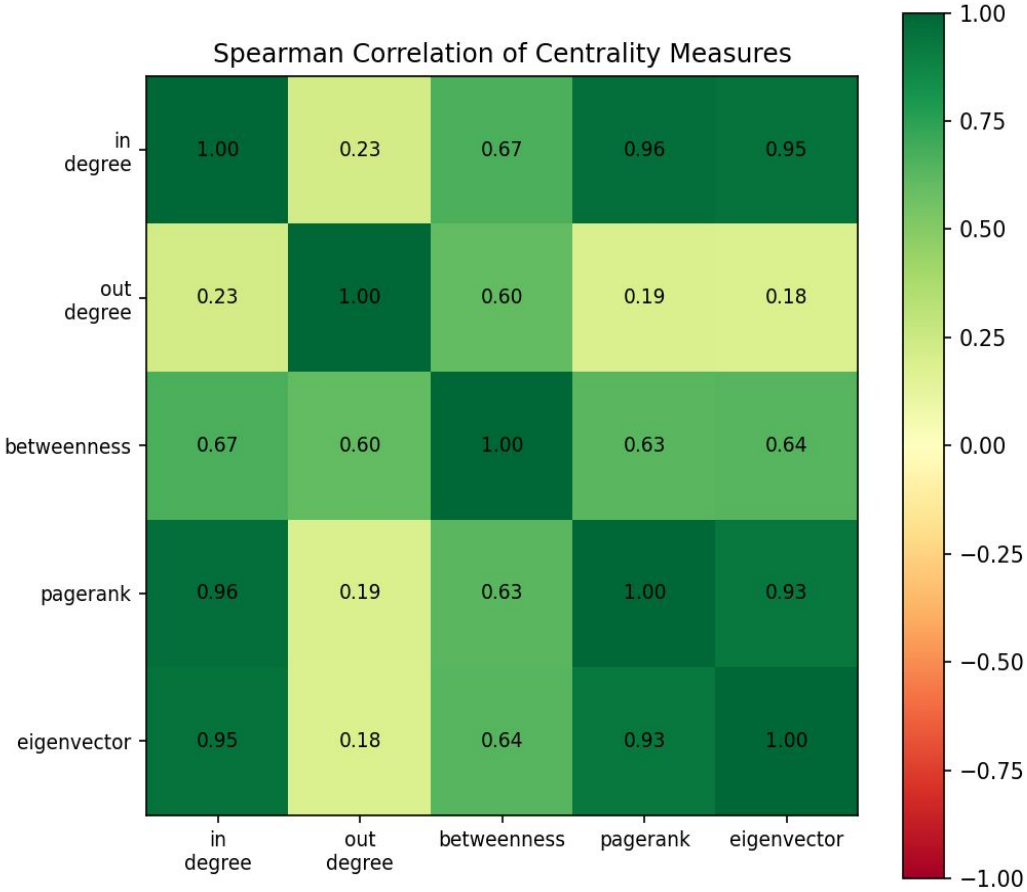
Measure	Captures	#1 Subreddit	Score
In - Degree	Popularity — linked TO	askreddit	0.081
Out - Degree	Activity — links OUT	bestof	0.046
Betweenness	Bridge between communities	subredditdrama	0.038
PageRank	Recursive importance	askreddit	0.022
EigenVector	Prestige — connected to important nodes	askreddit	0.224

Two distinct structural roles discovered:

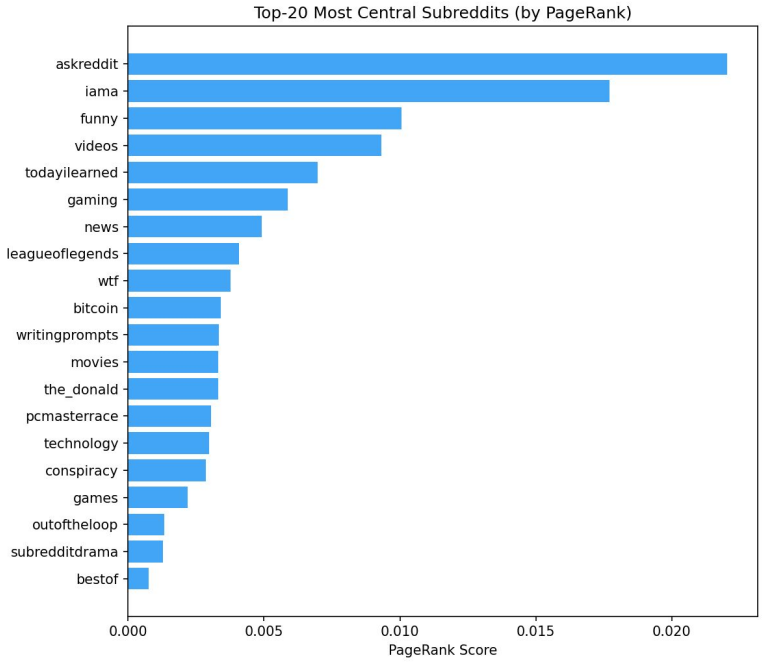
-Popular Hubs (In-degree ↔ PageRank ↔ Eigenvector, ρ = 0.93–0.96): askreddit, iama, pics, f funny
-Active Linkers (Out-degree, ρ ~ 0.2 with others): bestof, subredditdrama, titlegore

Consensus Top-5:

askreddit, iama, funny, gaming, todayilearned

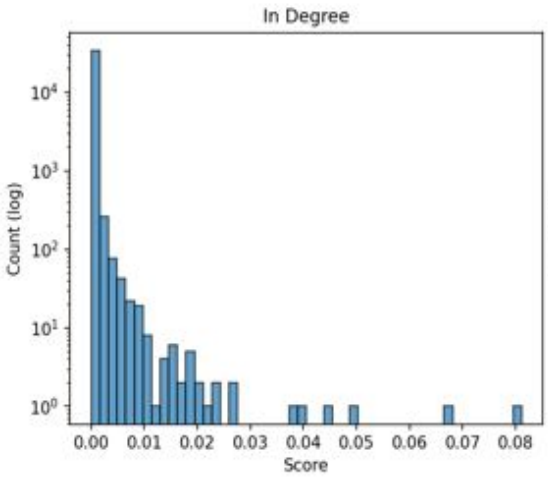


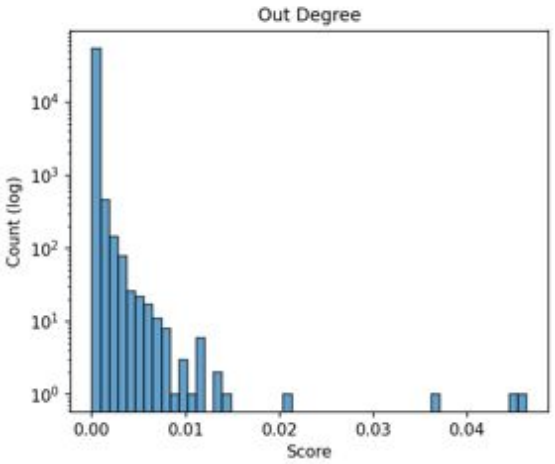
centrality_correlation

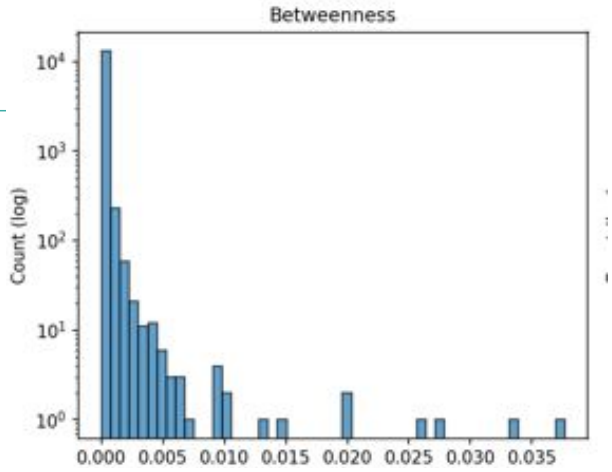


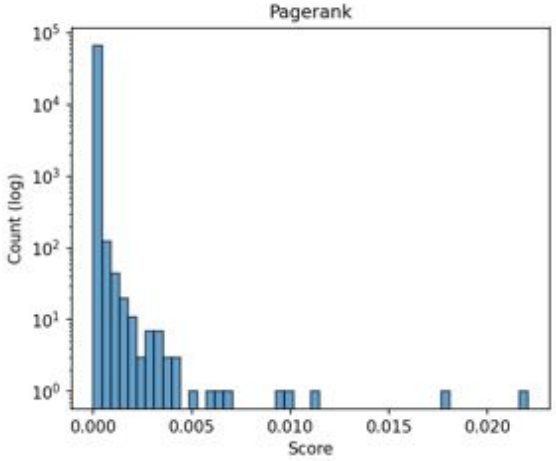
top_subreddits_pagerank

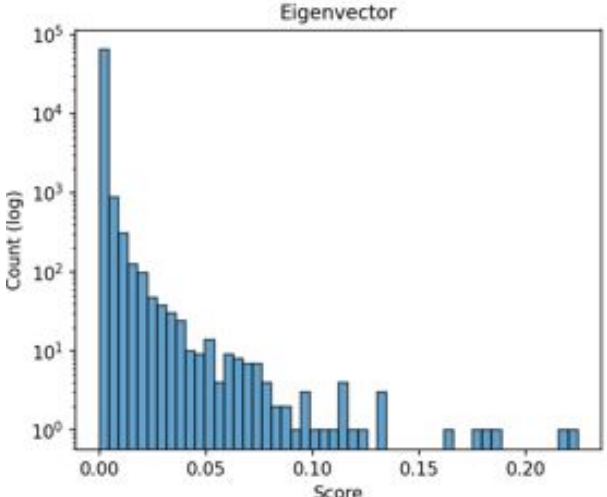
centrality_distributions











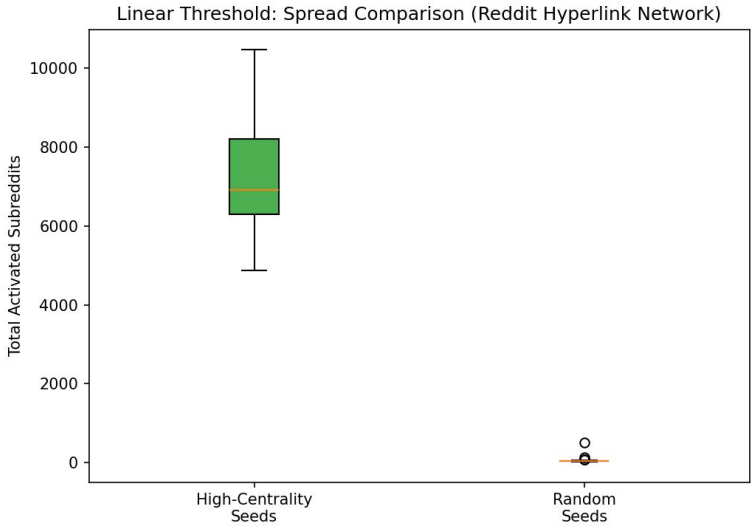
5) Information Diffusion Modeling

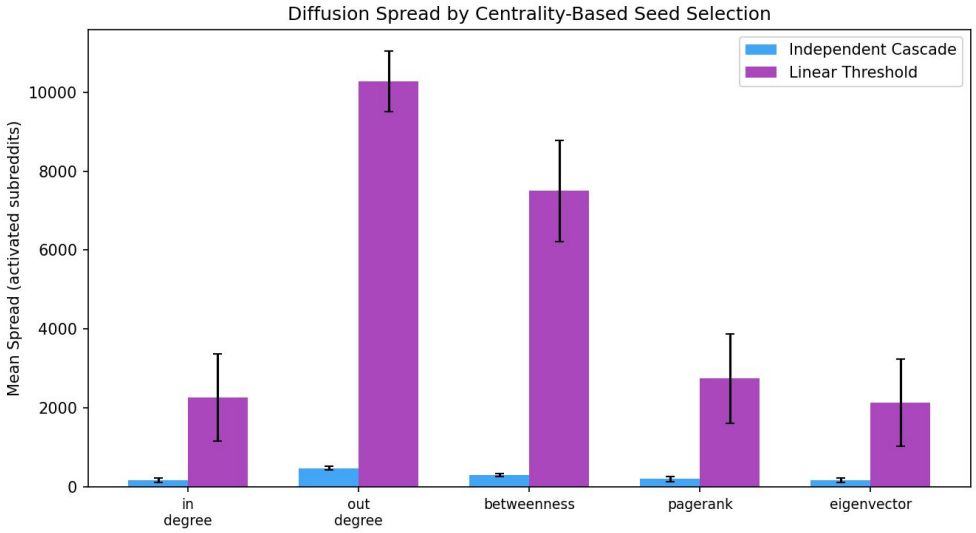
Two Models:

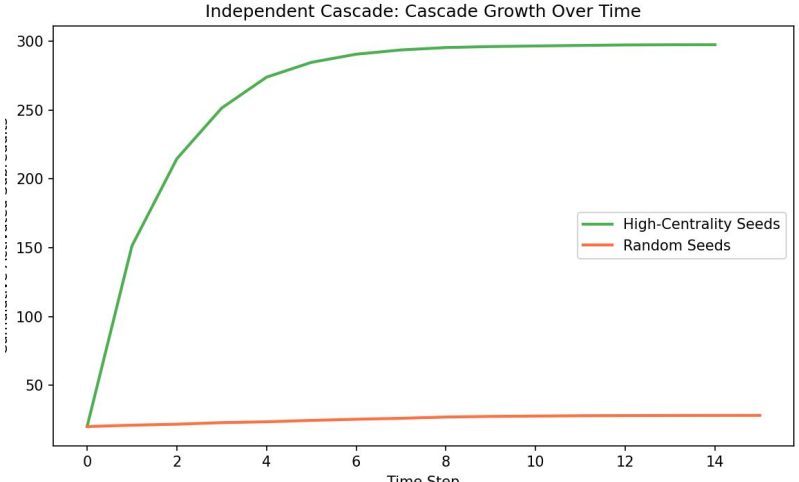
- Independent Cascade (IC): each active node tries to activate out-neighbors with p=0.01, one chance per edge (like a virus)
- Linear Threshold (LT): node activates when fraction of active in-neighbors exceeds random threshold (like peer pressure)

Model	Central Seeds	Random Seeds	Improvement
IC	302	29	+942%
LT	7,247 (10.8%)	53 (0.1%)	+13,583%

Centrality	IC Spread	LT Spread
Out-degree	473	10,278 ⭐
Betweenness	304	7,498
PageRank	206	2,749
In-degree	192	2,269
Eigenvector	162	2,132


Linear Threshold: Spread Comparison (Reddit Hyperlink Network)
Total Activated Subreddits
10000
8000
6000
4000
2000
0
High-Centrality Seeds
Random Seeds


Diffusion Spread by Centrality-Based Seed Selection
Mean Spread (activated subreddits)
Independent Cascade
Linear Threshold
10000
8000
6000
4000
2000
0
in degree
out degree
betweenness
pagerank
eigenvector


Independent Cascade: Cascade Growth Over Time
300
250
200
150
100
50
0
2
4
6
8
10
12
14
High-Centrality Seeds
Random Seeds

Cross-Validation and Conclusion

Measure	ρ	What it Predicts
Eigenvector	0.810 ⭐	Best predictor of receiving info
In-degree	0.745	Popular nodes are often reached
PageRank	0.709	Important nodes get reached
Betweenness	0.626	Bridges are moderately reachable
Out-degree	0.293	Active linkers are less reachable

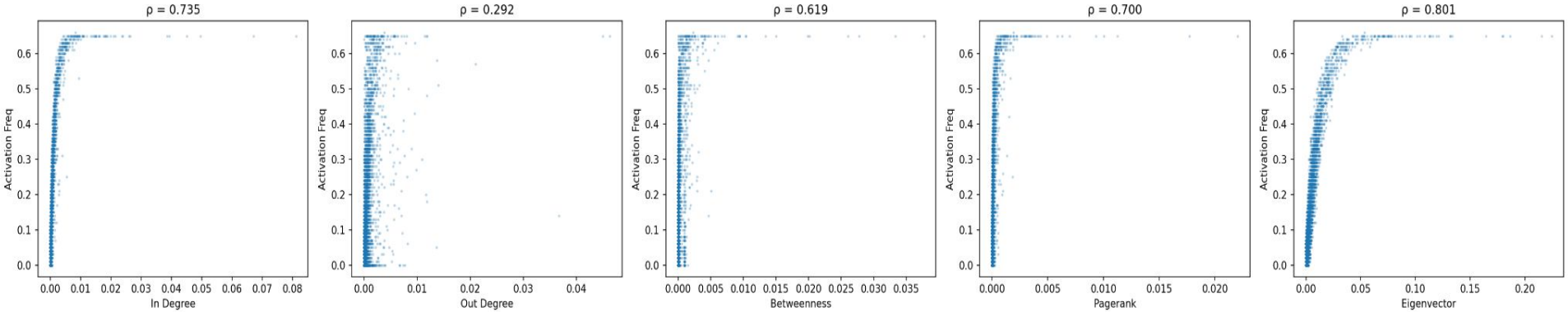
Two complementary findings from Parts 4 + 5:

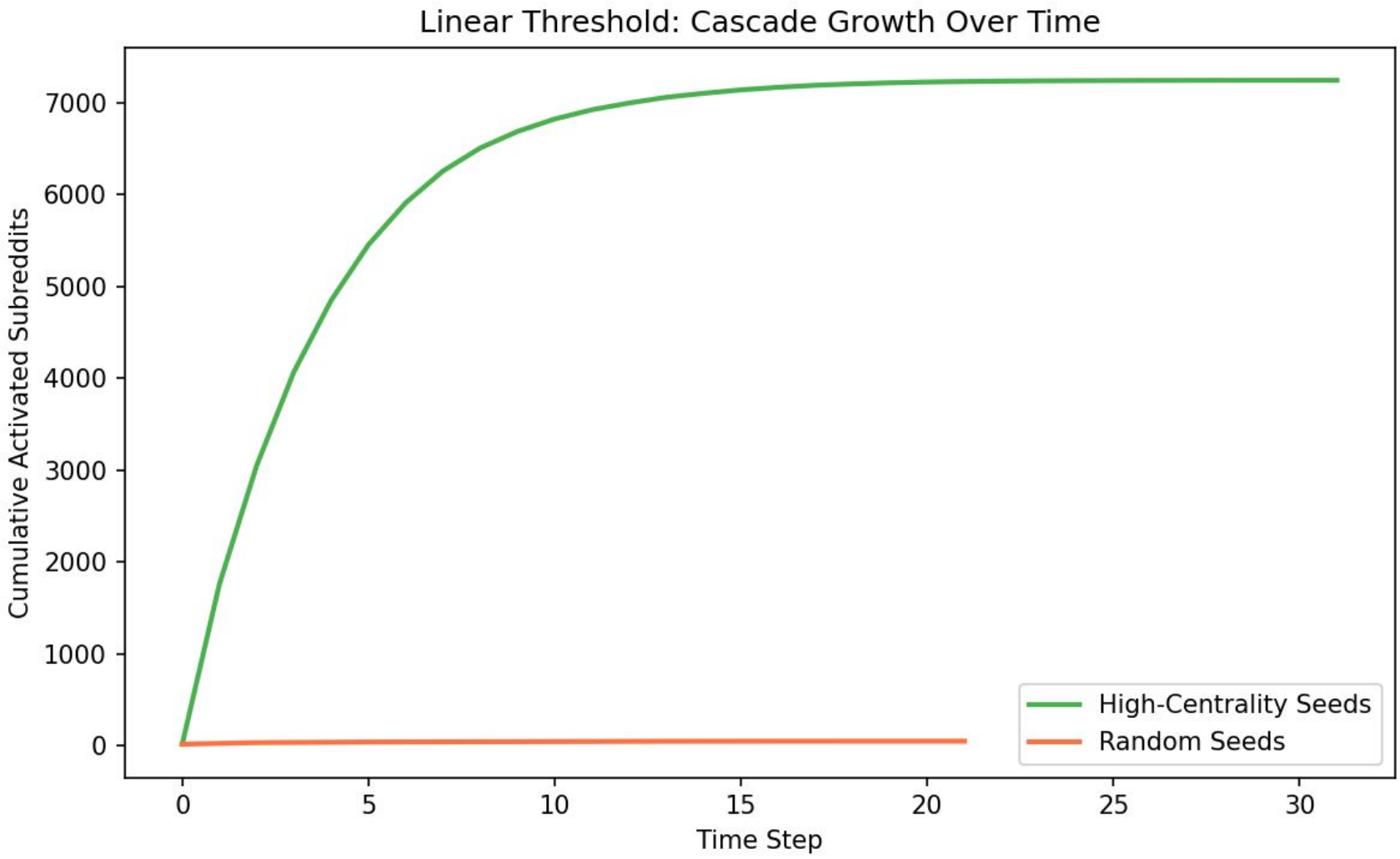
- Out-degree → best for SPREADING information (most outgoing channels)
- Eigenvector → best for predicting who RECEIVES information (ρ = 0.81)

Conclusion:

- Central seeds → 10x to 137x larger cascades than random
- Eigenvector centrality predicts diffusion participation at ρ = 0.81
- Static network structure reliably predicts dynamic information flow

cross_validation_centrality_diffusion




Linear Threshold: Cascade Growth Over Time
Cumulative Activated Subreddits
7000
6000
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
Time Step
High-Centrality Seeds
Random Seeds

6) Temporal Analysis and Prediction with TGN

- Beyond Static Graphs: TGNs treat networks as a continuous stream of events *(u, v, t)* rather than a frozen snapshot.

- The Memory Module: Each node maintains a state $s_i(t)$ that serves as a compressed history of all its past interactions.

- The Processing Pipeline:
  a. Message: Interaction occurs -> generate message from node states.
  b. Aggregate: Combine messages if multiple events occur simultaneously.
  c. Update: Use a GRU (Gated Recurrent Unit) to refresh node memory.
  d. Embedding: Apply temporal attention to neighbors to generate final predictions.

Training & Prediction Strategy

- Temporal Splitting: 70/15/15 split strictly by time to prevent looking into the future.

- Two Critical Tasks:
  - Transductive: Can we predict links between known nodes?
  - Inductive: Can we generalize to entirely new nodes never seen in training?

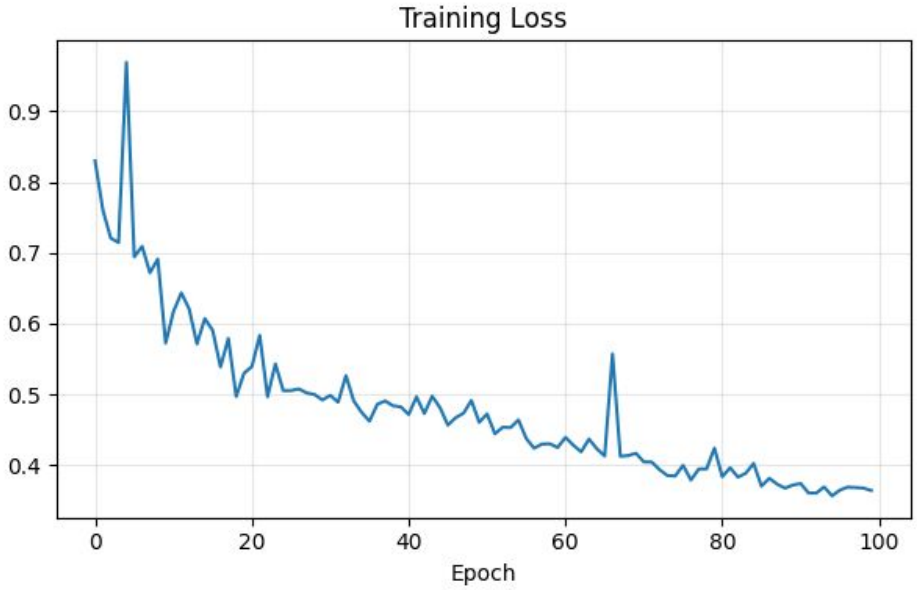
- The Learning Objective: Trained via Binary Cross-Entropy. The model must distinguish a real interaction *(u, v, t)* from a randomly sampled fake pair *(u, w, t)*.

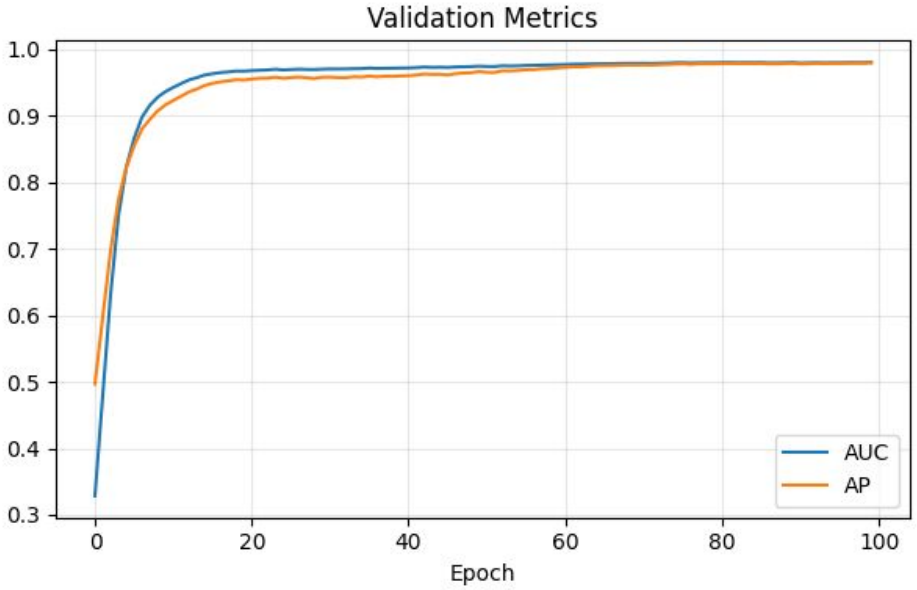
Results & Metrics

Test Set Statistics:

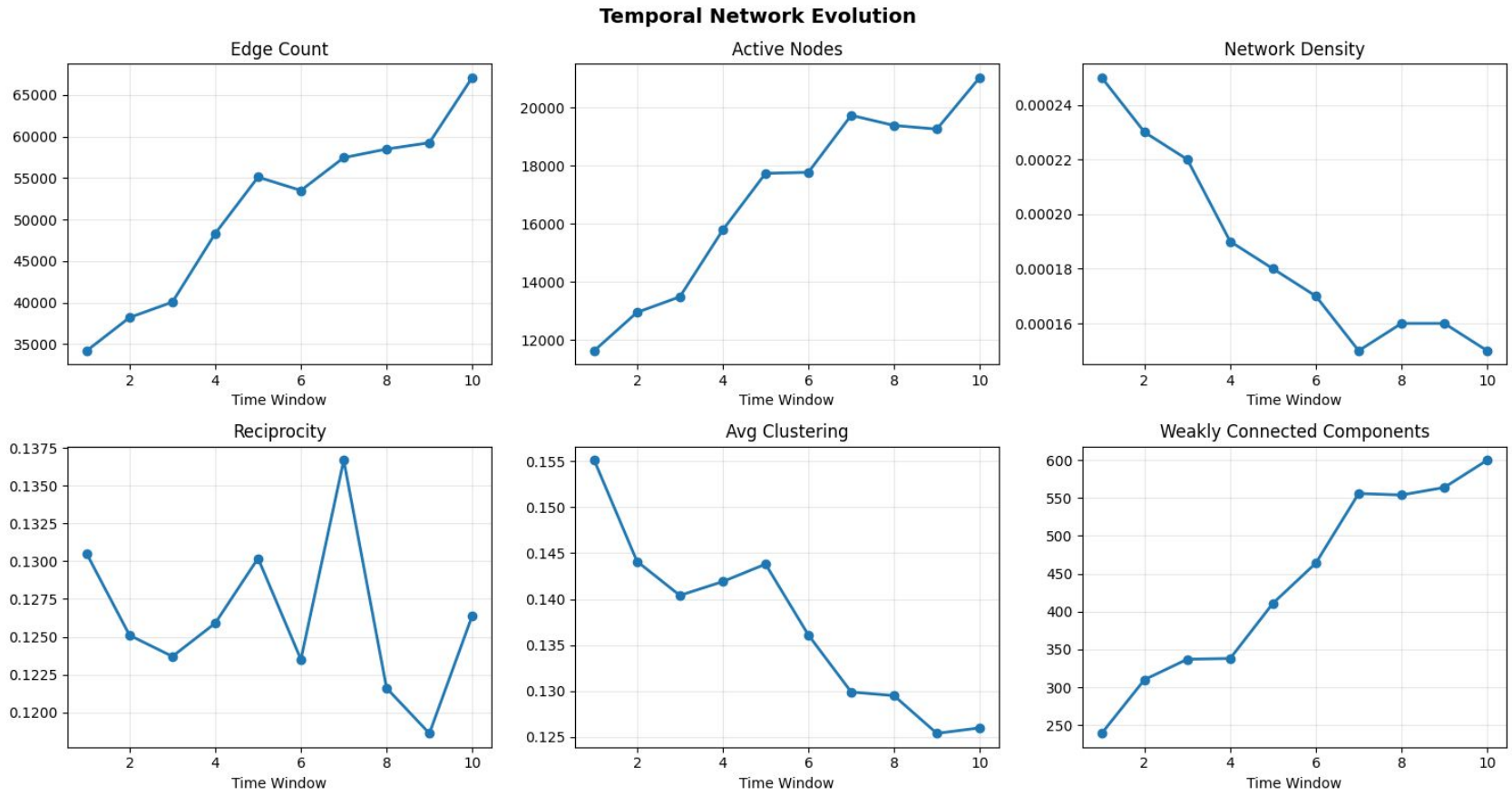
AUC-ROC : 0.9766

Average Precision: 0.9756





Longitudinal Network Analysis



THANK YOU

OPEN TO Q/A

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